Crown and bridge

Partial veneer crown preparation

Lecture 8

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Three quarter crown:

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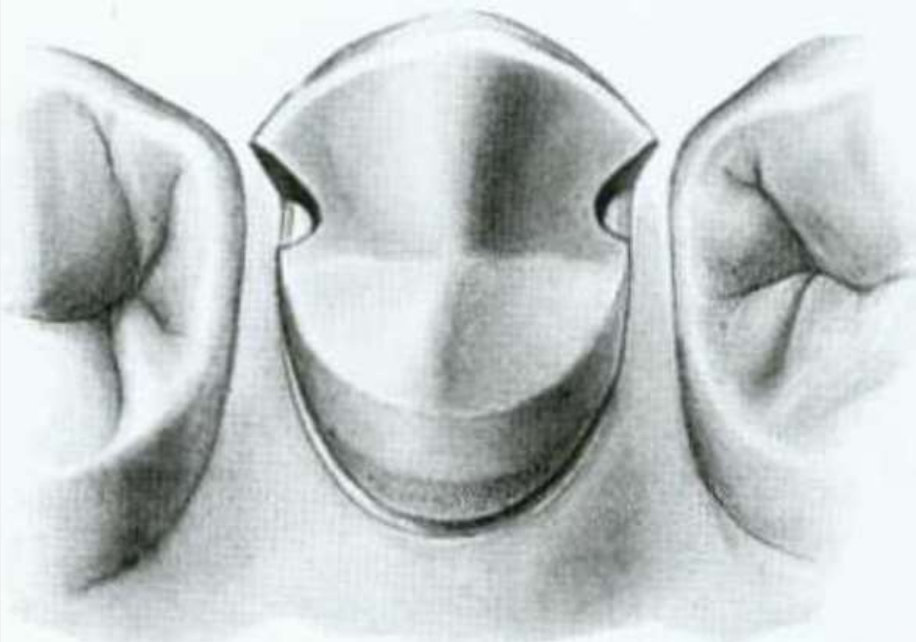
It is a cast metal restoration which covers 3/4 of the crown (occlusal or incisal, lingual, and proximal surfaces) leaving the labial or buccal surface unprepared. It is less retentive and less resistant to displacement compared to full metal and full veneer crown.

The resistance to lateral forces or buccolingual displacement of the restoration is prevented by internal features like proximal boxes or grooves)

Uses:

1-As a retainer for short span bridge.
2-As a single restoration.
3-As a splint in anterior teeth.



INDICATIONS

1-On teeth with clinical crown of good length and thickness labiolingually.

2-Patient with good oral hygiene.

3-On teeth without or with minimal caries on buccal or labial surface.

4-No discrepancy in axial relationships of the tooth and path of insertion of a bridge (good alignment of the prepared tooth with other teeth) .

INDICATIONS

For posterior teeth:

1. Lost moderate amount of tooth structure with intact and well supported buccal surface.

2. Retainer for fixed partial denture.

For anterior teeth;

1. Suitable for teeth with a sufficient bulk and intact labial surface.
2. 2. Retainer for F.P.D. or splinting of anterior teeth.

Contraindications

1-Short teeth.

2-Poor oral hygiene.

3-Narrow proximal surfaces.

4-Tooth with extensive caries.

5-Long span bridge.

6-Non vital teeth.

7- Poor alignment.

Advantages of 3/4 crown:

1-Tooth structure is saved

2-Better esthetic than other types of crowns.

3-Vitality test can be done on the unprepared tooth surface.

4-Less chance of periodontal irritation because all the margins of the crown is supragingival

5-Complete seating of the crown can be easily seen by direct observation.

Disadvantages:

1-Difficulty in preparation compared to other types of crowns.

2-There is more possibility of recurrent caries along the crown margins (longer margins).

3-There is a possibility of showing metal especially in the lower anterior and posterior teeth.

4-Less retention and resistance than complete cast crown.

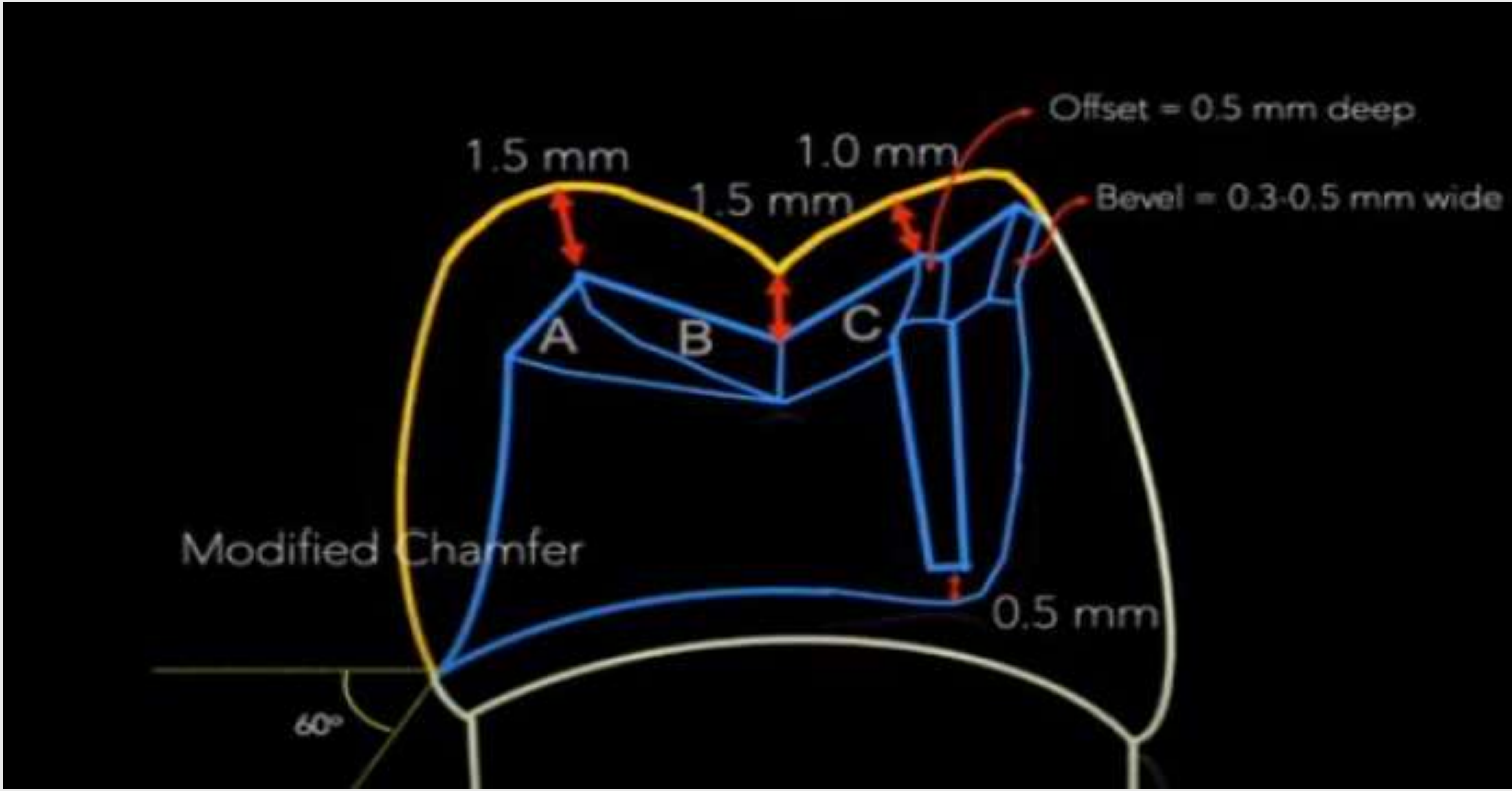
5-Limited adjustment can be done in the path of insertion.

Tooth Preparation :

Recommended dimensions:

- 1.5 mm on functional cusp (lingual).
- 1.0 mm on non-functional cusp (facial).
- Less than 0.5 mm on facial cusp tip if sufficient horizontal overlap.
- 1.5 mm clearance.
- Follow contours of opposing tooth.
- Maintain contours of tooth being prepared.

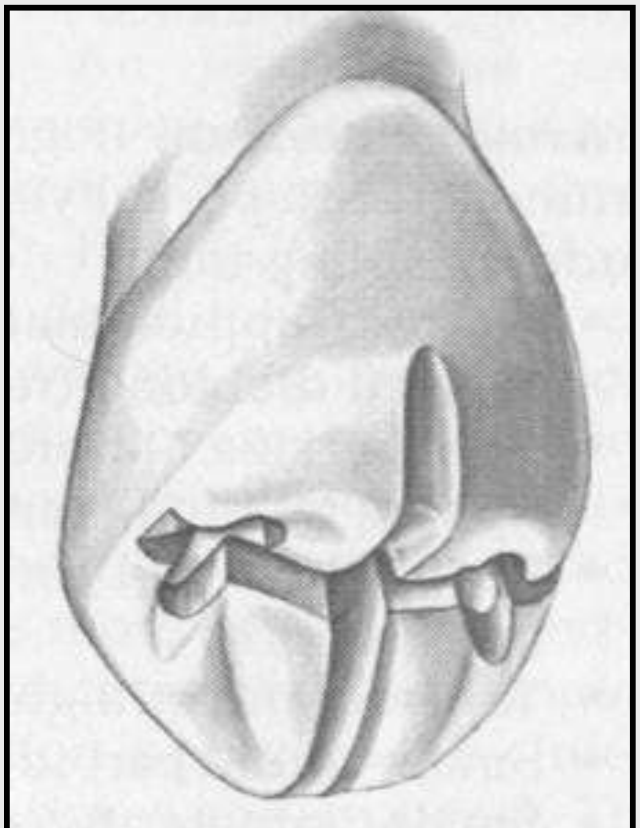
Recommended dimensions

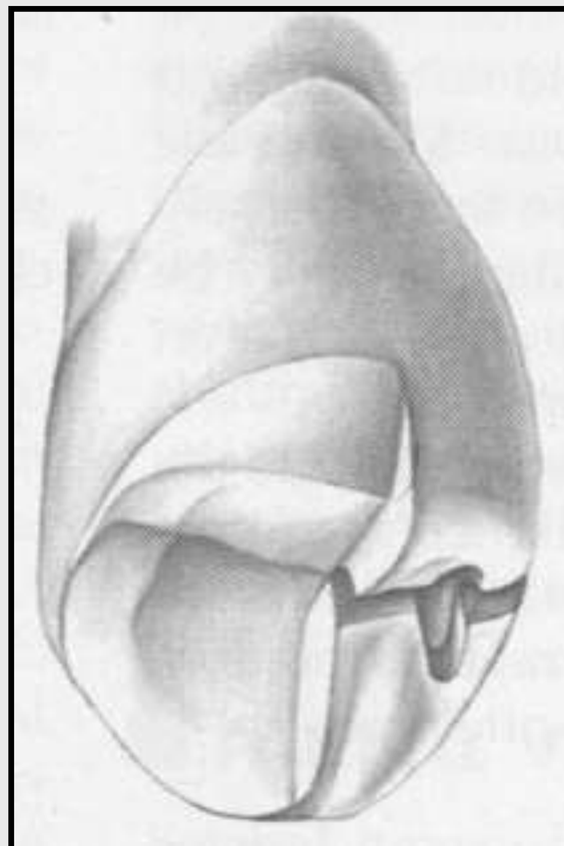


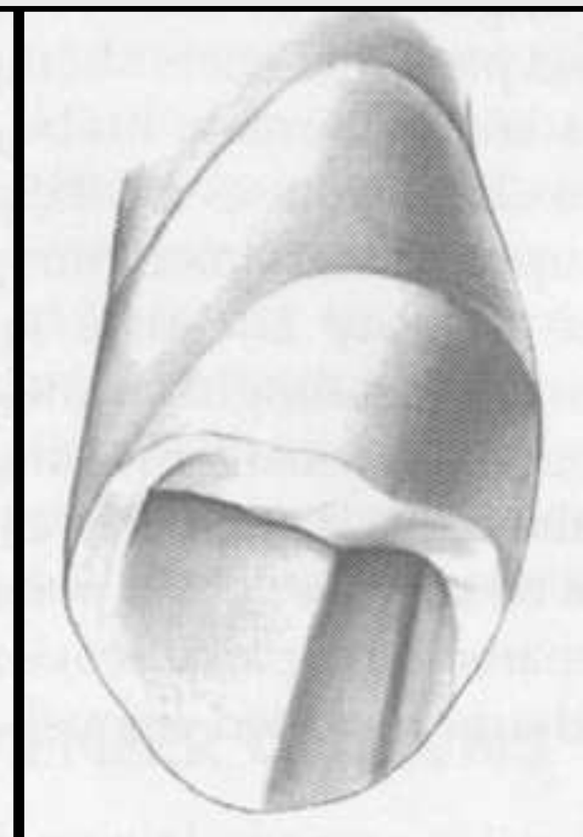
Occlusal Reduction.

Initial depth holes are placed in the mesial and distal fossae approximately 0.8 mm deep.

Three guiding grooves in each cusp and for functional cusp for bevel similar to that of complete cast crown.







1.Occlusal surface preparation:

1. D.O.G. placed on the anatomic ridge and grooves of occlusal surface using round end taper fissure bur, the grooves should extend through occluso-buccal line angle but only with 0.5mm deep to prevent metal display.

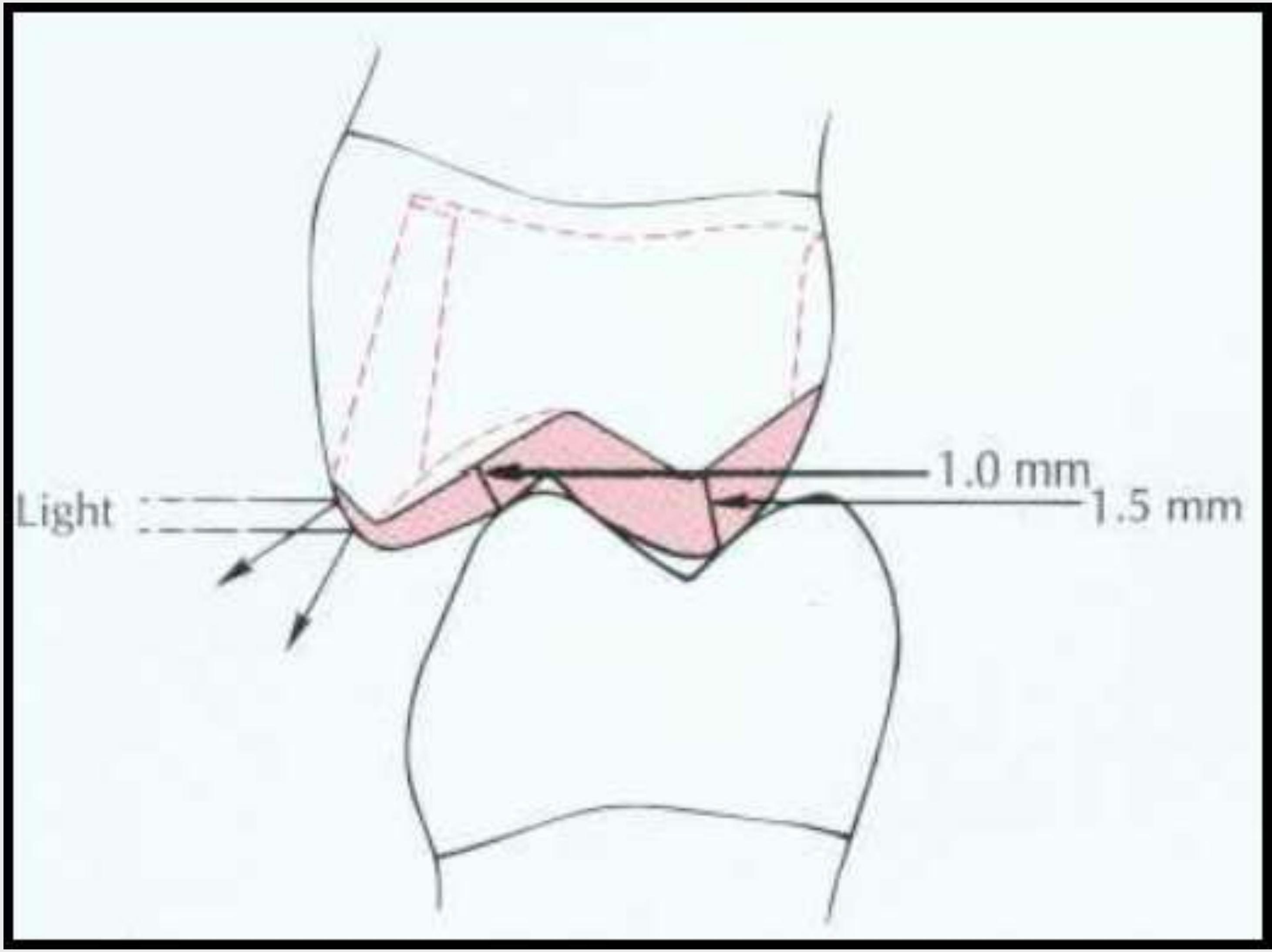
2.Occlusal reduction were then complete by removing tooth structure between grooves reproducing the geometric inclined plan pattern of cusps, the depth of reduction should be decrease at the occlusobuccal line-angle.

3.Awide bevel is placed on the functional cusps using the same bur .

4.Occlusal clearance were then check in centric & eccentric relations

The exception in the groove application is the lingual inclination of the buccal cusp. It becomes shallower as they approach the buccal cusp tip.

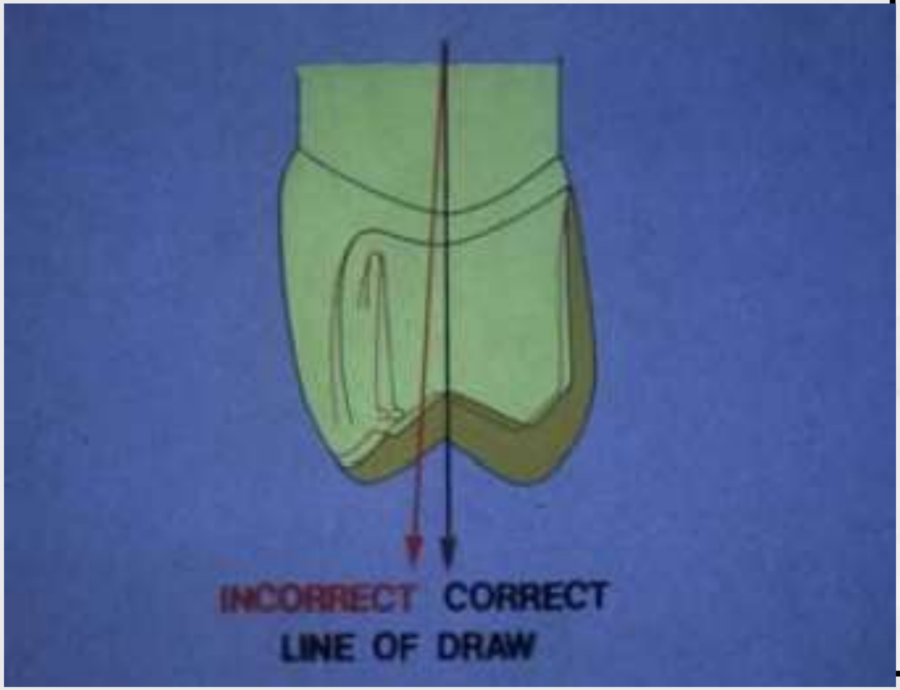
Upon the completion of occlusal reduction, a clearance of at least 1.5 mm should exist on the centric cusp and at least 1.0 mm on the eccentric cusp and in the central groove.

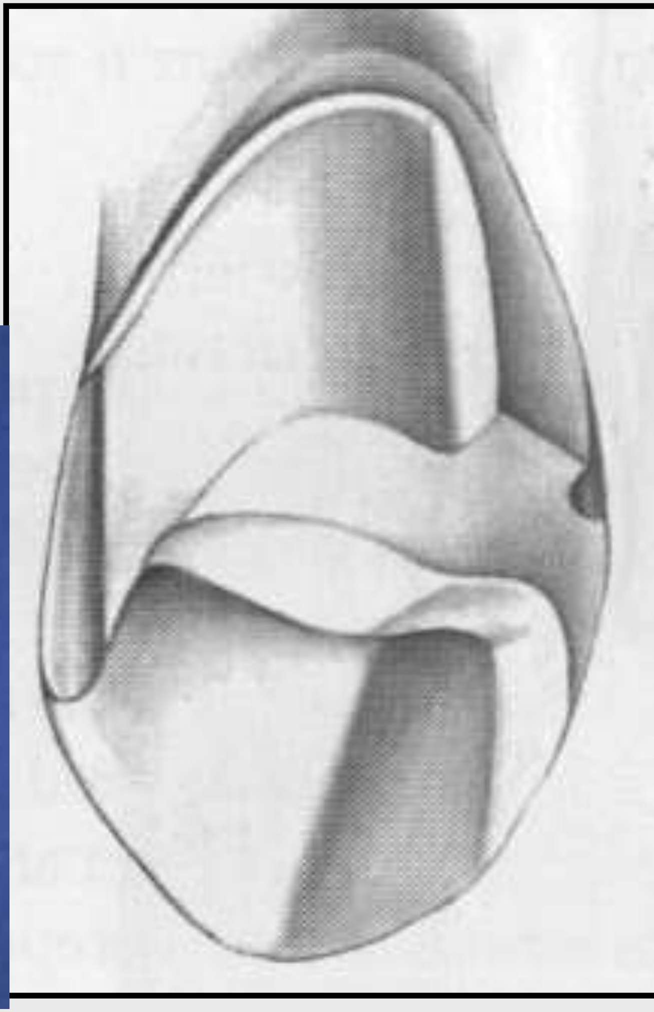


Lingual reduction:

Three grooves should be parallel to the long axis of the tooth and should not exceed half the width of the diamond fissure bur.

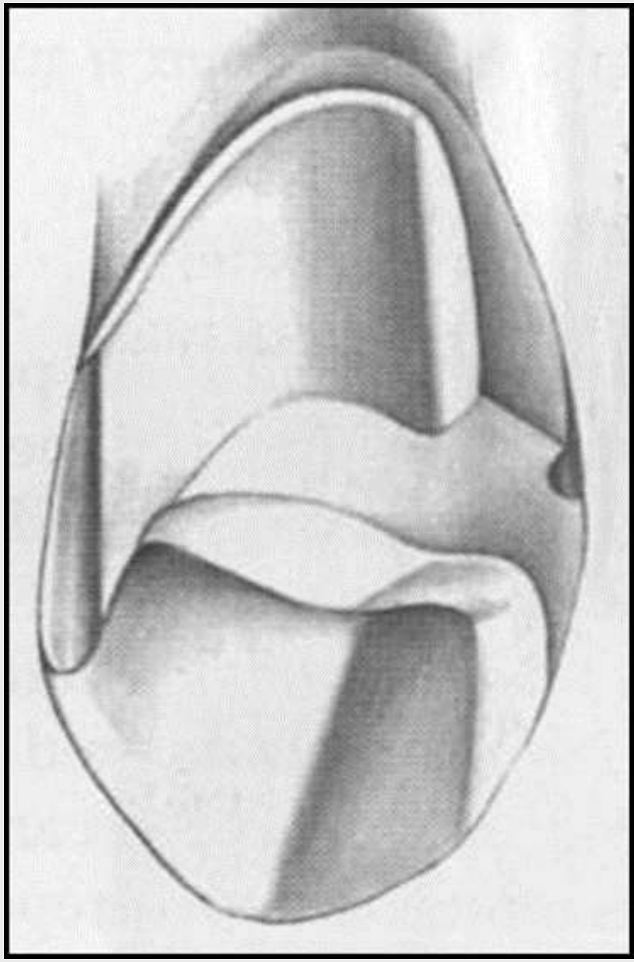
A common mistake to incline the groove toward the buccal. This will reduce the retention. But can be corrected. Remove tooth structure between the grooves and place a cervical chamfer.





Interproximal Reduction:

Proximal access is gained by short needle diamond, up and down movement, this continue until contact opened slightly short of breaking the proximal contact,



The resulting flange should be parallel to the lingual preparation, with the chamfer placed sufficiently cervical to provide at least 0.6 mm of clearance with the adjacent tooth.

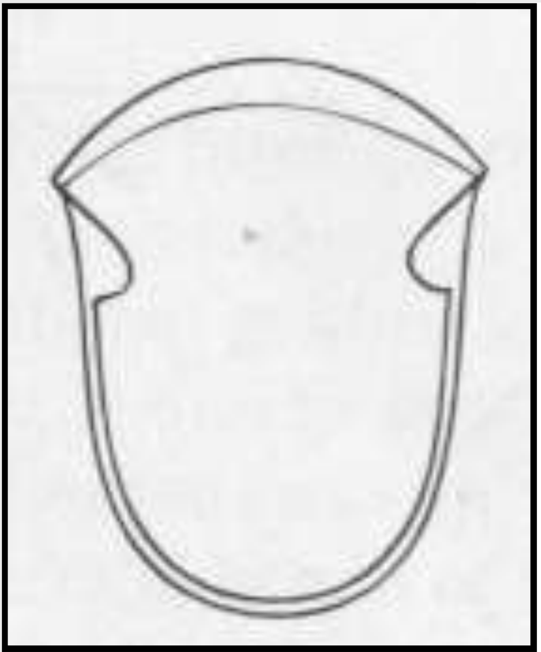
The axial wall allowing for a proximal groove of at least 4mm length occlusocervically.

Avoid damage to adjacent tooth and excessive axial reduction.

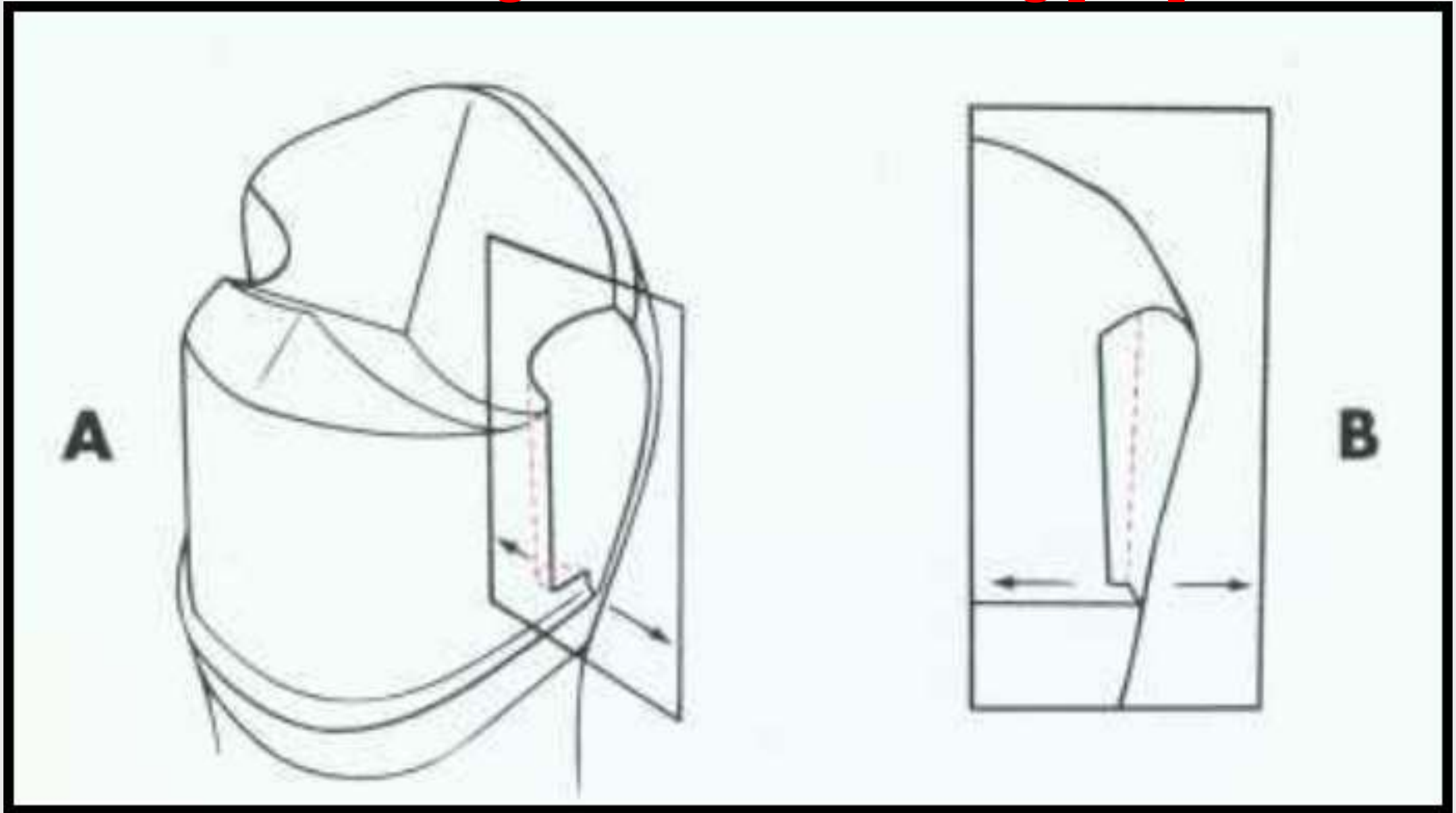
Retention groove Placement:-

Proximal grooves (mesial and distal) are placed parallel to the path of insertion and parallel to each other using carbide fissure bur. Normally, unsupported tooth structure will remain on the buccal side, and this side is flared to remove it.

Position the bur against the interproximal flange of the prepared surface parallel to the long axis of the tooth.



Enter full depth of the bur, the groove need not to be deeper than 1mm at its cervical end but may be deeper near its occlusal end. This is because of the tapering of the carbide bur. Cervical end of the groove should be above the chamfer finishing line for burnishing purposes.

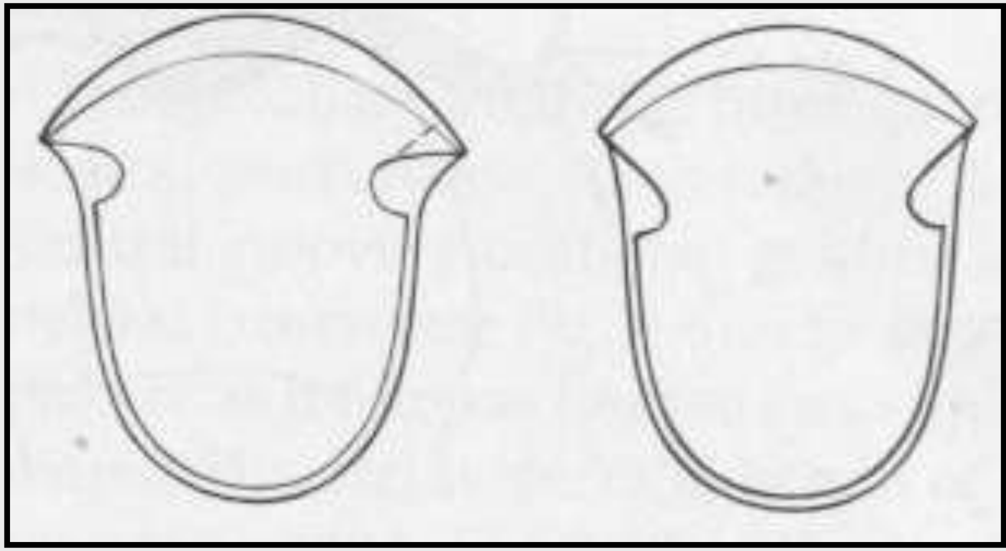


<u>Criteria of the proximal Groove:-</u>

The groove should resist lingual displacement of an explorer, the lingual walls of the proximal groove should have 90-degree angle, to resist lingual displacement.

The walls of the grooves should not have an undercut relative to the long axis of the tooth.

The walls should be flared toward the intact buccal surface of the tooth. <u>The aim</u> of flaring is removing any unsupported tooth structure remain at the cavosurface angle of the buccal surface.



Summery of requirements of retentive grooves:

1. It should cut to full diameter of carbide bur No.171(0.5mm) to create defiant lingual wall.
2. It should extend to the full length of proximal wall (ending about 0.5mm to the chamfer).
3. It should be placed as far as facially as possible without undermining facial surface (between middle and labial third).
4. It should be parallel to the long axis of the tooth.

Advantages of Proximal grooves:

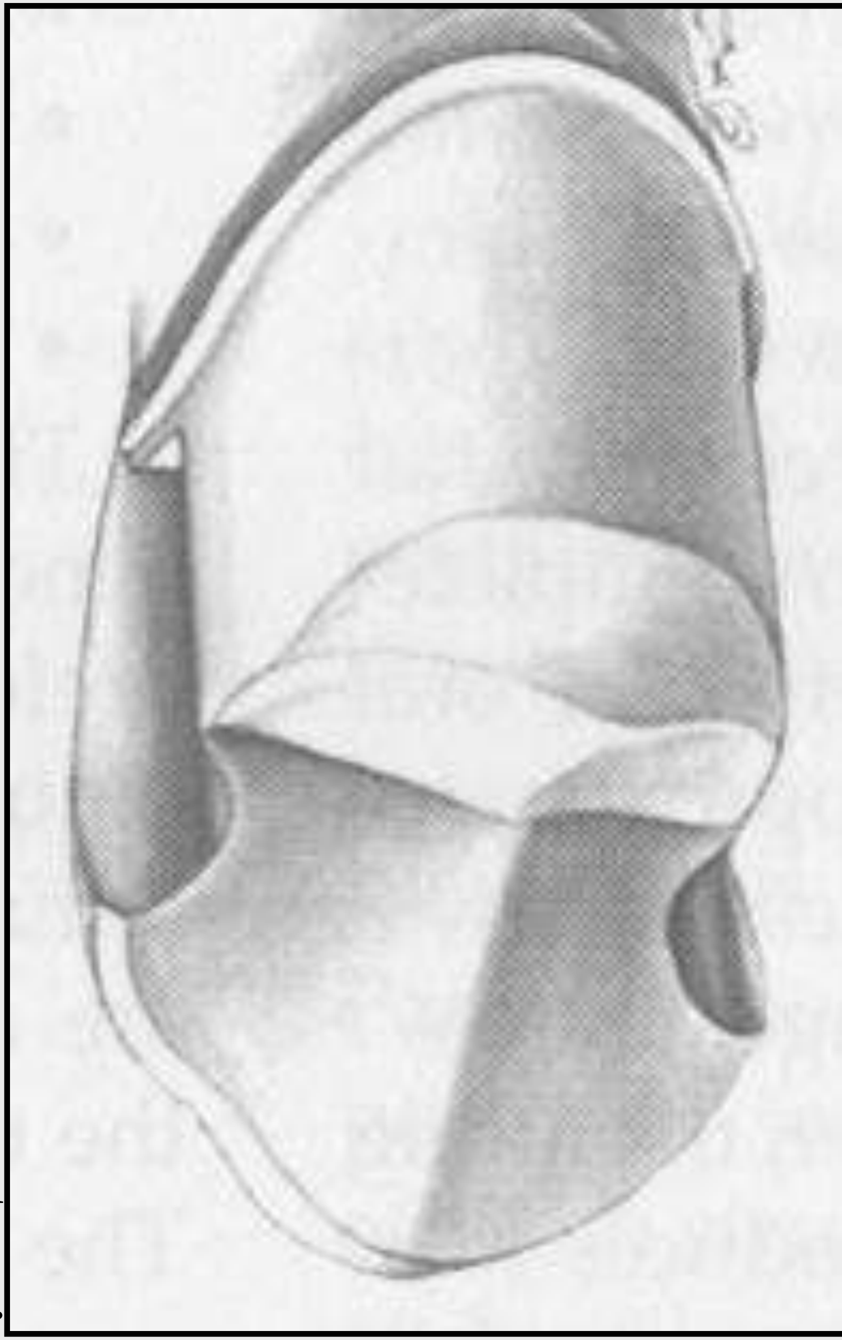
1. Increase retention.
2. Prevent rotation (resistance).
3. Reinforce the margin of restoration at this area.
4. They act as a guide during placement.

<u>Buccocclusal Contrabevel:-</u>

It is a narrow contrabevel that connects the mesial and distal Flares.

Purpose: - To remove any unsupported enamel and protect the buccal cusp tip from chipping during function.

Criteria: - The bevel should remain within the curvature of the cusp tip rather than extending it to the buccal wall. Thus the restoration will be less obvious.

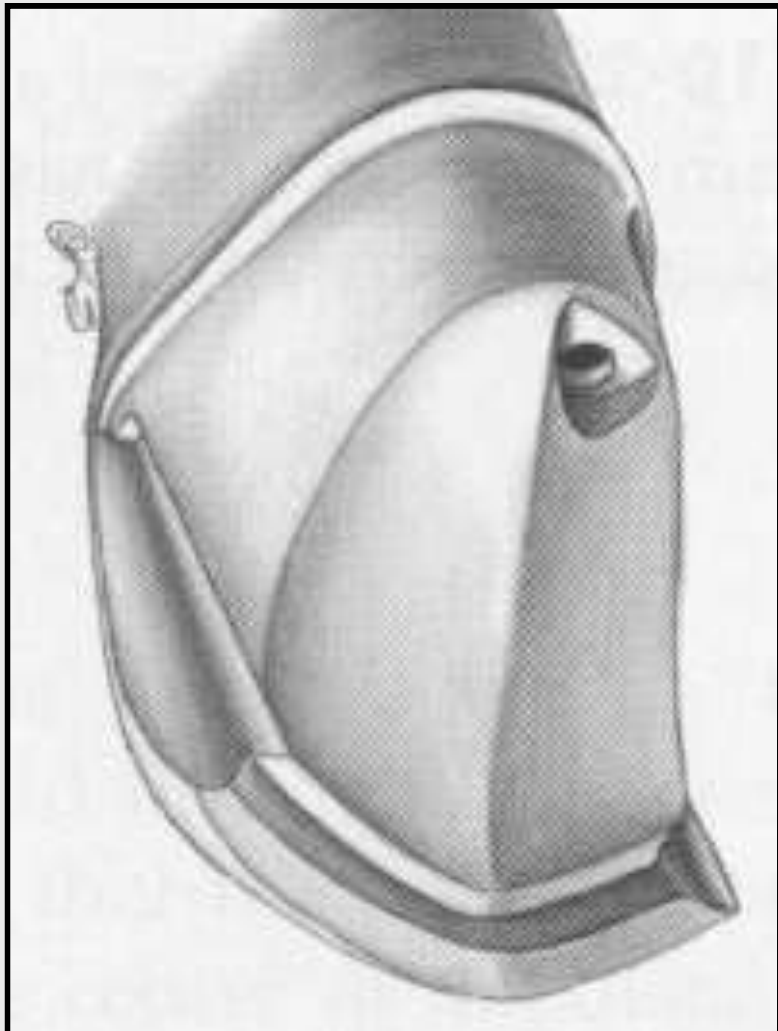


Occlusal offset:-

It is a V-shape groove extends from the proximal grooves along the buccal cusp. The inverted cone diamond or carbide can be used.

Purpose: - Give additional bulk to ensure rigidity of the restoration.

Indications: - It is not usually necessary for posterior crowns. but is essential for the structural durability of anterior crowns.



Special consideration:-

Avoid extending the cervical chamfer closer to the cementoenamel junction. Because more axial tooth structure will be removed.

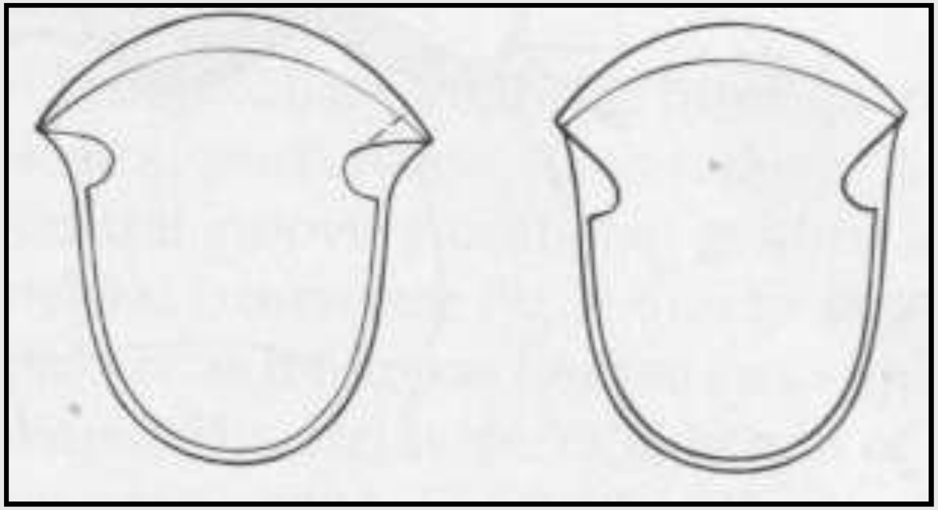
Finishing:-

A - Round all sharp internal line angles.

B – Reevaluates :-

1- Flares.

2- Any remaining undercuts.



The mesial flare cannot extend beyond the transitional (mesio buccal) line angle for esthetic purposes, while in distal aspect; flare may extend slightly farther to the buccal, allowing better access for oral hygiene (self-cleansing area).

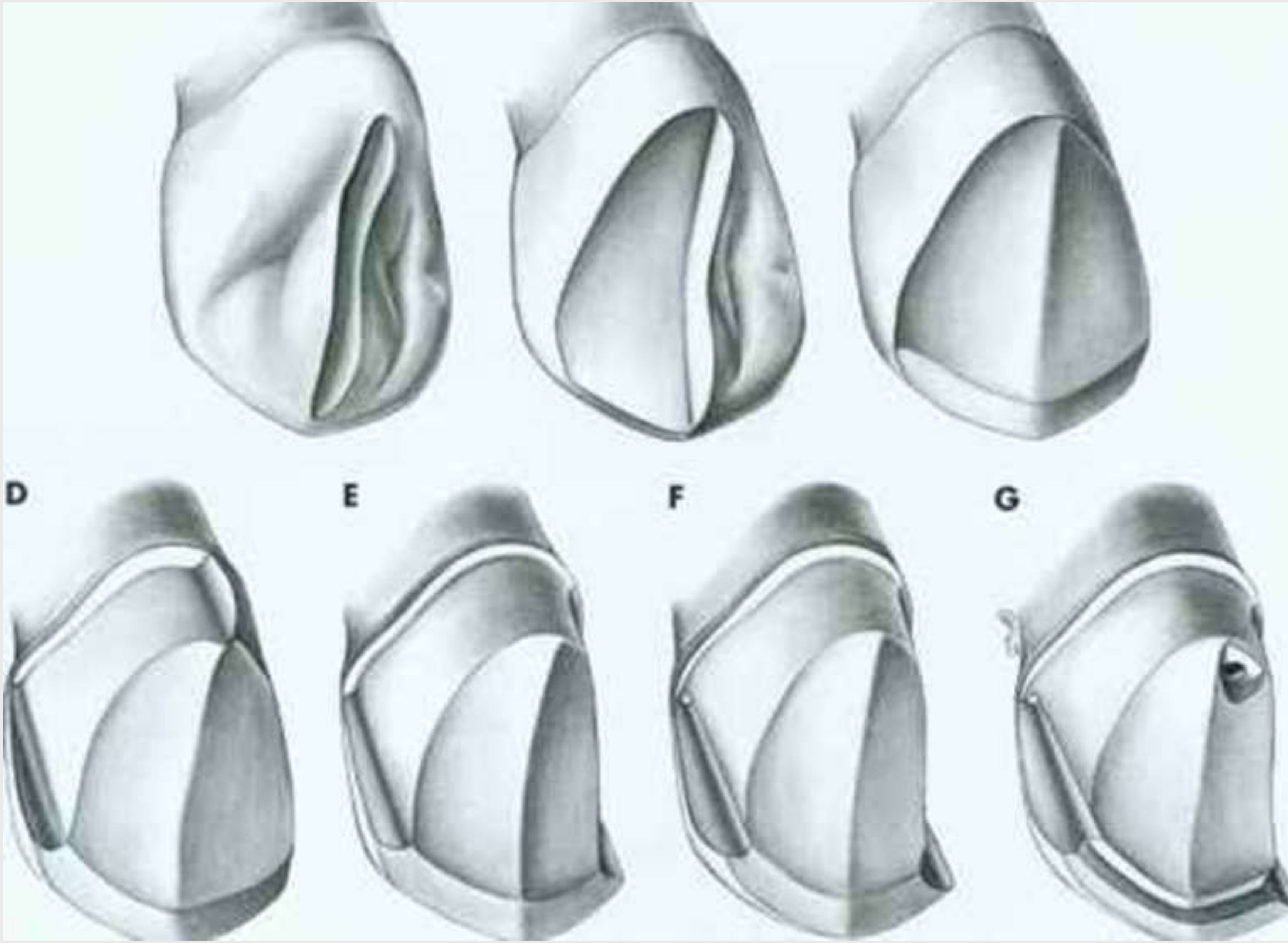
3/4 crown preparation on anterior teeth

lingual surface:

This is done by two-step reduction similar to other types of crown preparations.

1-cingulum portion.

2- lingual fossa portion.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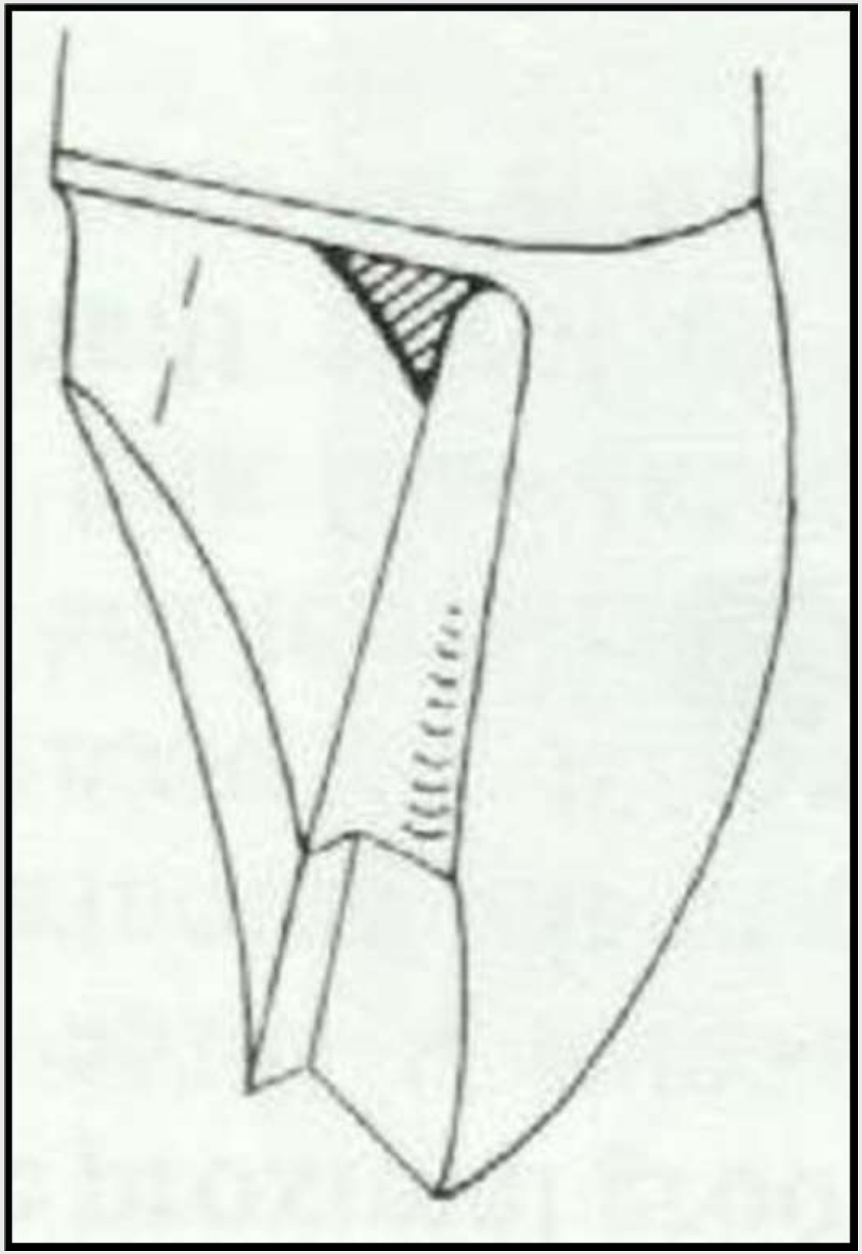
Incisal reduction :

For the maxillary anterior teeth a lingual-incisal bevel is placed using a diamond bur at 45°to the path of insertion, this preparation should not be extended labially to prevent appearance of metal.



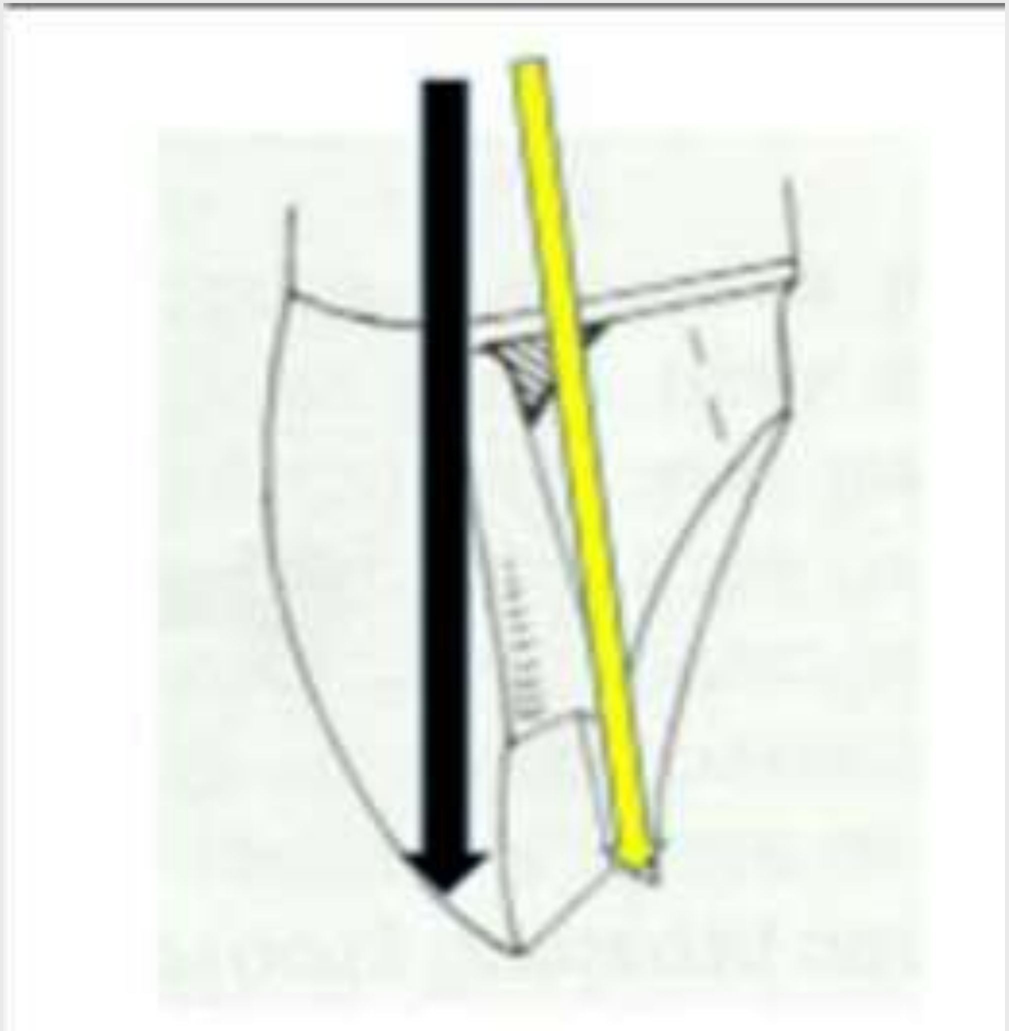
For the lower anterior teeth, a reverse bevel is placed on the labial surface to cover the incisal edge in order to:

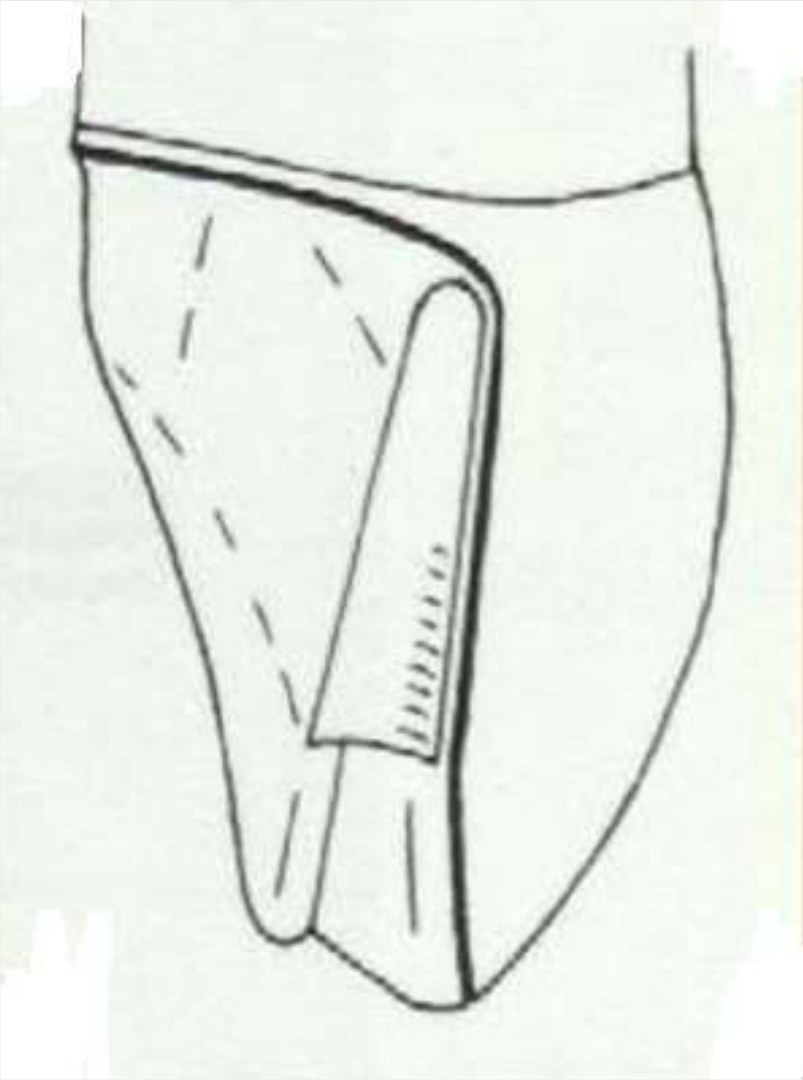
1- protect the area of unsupported enamel from fracture

2- prevent the dislodgement of the crown to the lingual direction.

Proximal reduction:

- **The area is prepared similar to the full metal with facing crown but care should be taken to avoid extending the reduction to the labial surface.**
- **Two proximal grooves should be placed parallel to the incisal 2/3 of the labial surface using a carbide fissure bur to get the longest groove for better retention and to avoid overcutting to the labial surface resulting in poor esthetic.**
- **The grooves should be placed at the junction of the labial and middle third of the proximal surface and parallel to each other.**
- **The base of the groove should be 0.5 mm. above the chamfer finishing line.**
- **The mesial and distal grooves should be connected with an incisal offset.**
- **The advantage of the incisal offset is the improvement of the strength of casting at this area and reinforcement of margin by connecting the two proximal grooves together.**

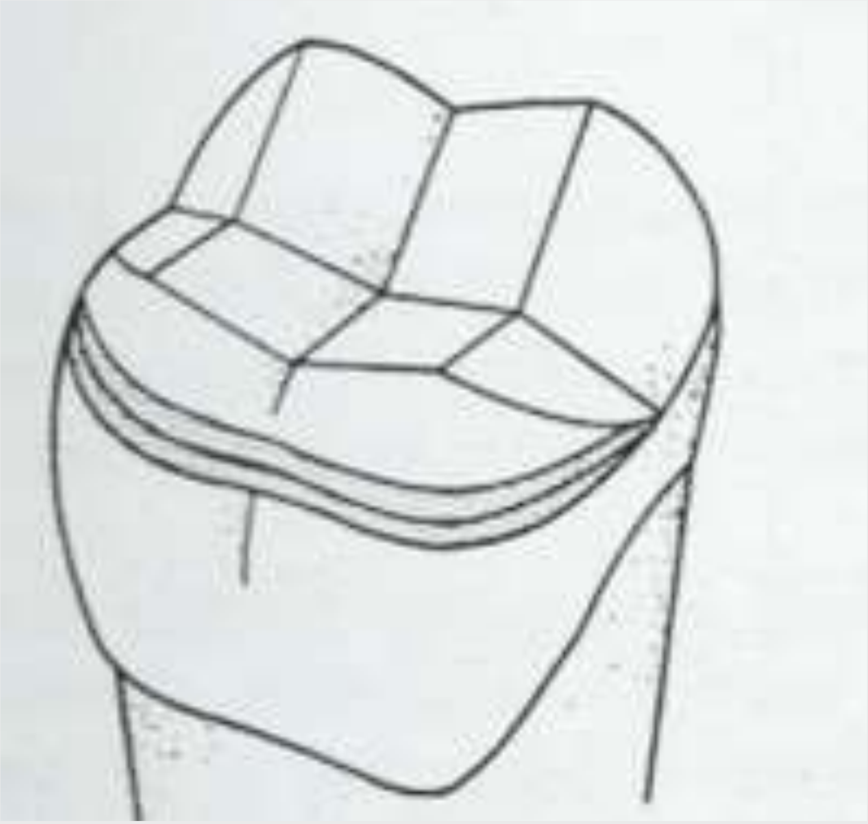




Differences in the preparation of upper and lower posterior teeth

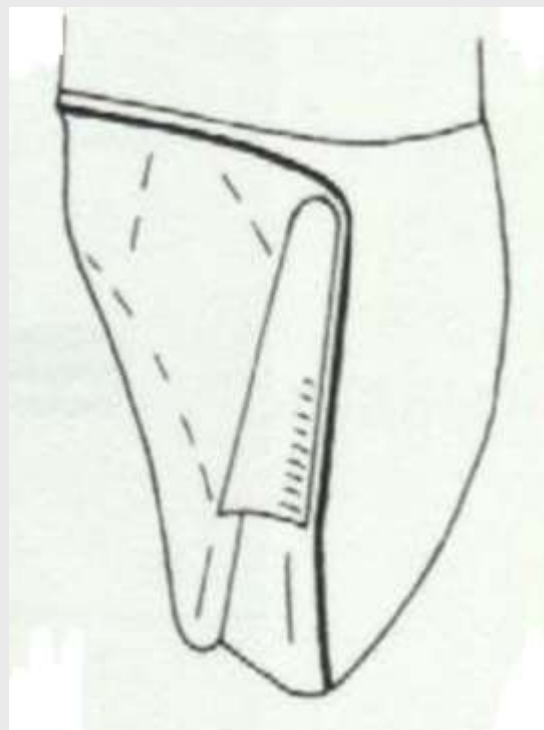
1-The occlusal finishing line of the maxillary teeth terminate near the bucco-occlusal junction while in the lower teeth the finishing line is 1 mm. gingival to the lowest occlusal contact with the upper teeth because the buccal cusp in lower teeth is a functional cusp which should be covered by metal.

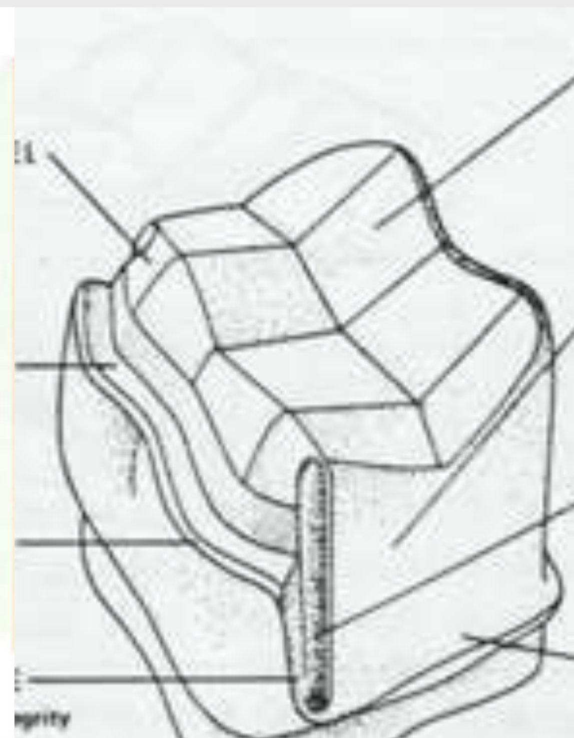
2-In the maxillary teeth there should be an offset while in the lower teeth there is no offset but there is a bucco-occlusal finishing line (more reduction).



Differences between anterior and posterior teeth preparation

In the anterior teeth the retentive proximal groove should be parallel to the incisal 2/3 of the labial surface while in the posterior teeth it is parallel to long axis to get the longest groove for better retention of crown.



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